

Stage 1: NaCo-DS

**Client
Agent**



**Counselor
Agent**

*Persona-Persistent
Dialogue Synthesis*

Stage 2: CoRA-MCTS

Select

Expand

Simulate

Backprop

*CoT Reasoning + RAG
Multi-dim Reward*

Stage 3: Prog-SimPO

**Turn-level
Pref. Pairs**



**Progressive
Alignment**

*Length-Normalized
SimPO Loss*

NaCo: Three-Stage Pipeline for Naturalistic Counseling